



ក្រសួងអប់រំ យុវជន និងកីឡា
វិទ្យាស្ថានបច្ចេកវិទ្យាភ្នំពេញ



ជេហ៍តីម៉ង់គណិតវិទ្យាអនុវត្តន៍ និងស្ថិតិ

គម្រោងសញ្ញាប័ត្រវិស្វកម្ម

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DE LA JEUNESSE ET DES SPORTS**

INSTITUT DE TECHNOLOGIE DU CAMBODGE
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ក្រសួងអប់រំ យុវជន និងកីឡា
វិទ្យាស្ថានបច្ចេកវិទ្យាកម្ពុជា



ដេប៉ាតឺម៉ង់គណិតវិទ្យាអនុវត្តន៍ និងស្ថិតិ

គម្រោងសញ្ញាប័ត្រវិស្វកម្ម

របស់និស្សិត: តាំងគីងម៉េង

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: បណ្ឌិត ភោគ សុខឌី

សាស្ត្រាចារ្យណែនាំ

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អ្នកទទួលខុសត្រូវក្នុងសហគ្រាស : លោក

រាជធានីភ្នំពេញ, ២០២៦

INTERNSHIP REPORT ON

“”

at

Submitted By:

TAING KORNGMENG

Under the supervision of:

Mr

Under the advisory of:

Mr. CHHEANG VINHA

Defense date:

In partial fulfillment of the requirements for the Internship Program

for the award of the degree of

Engineering in Applied Mathematics and Statistics

Majoring in Data Science

The logo for the Department of Applied Mathematics and Statistics (AIIS) features the letters 'AIIS' in a bold, dark blue serif font. A vertical red bar is positioned between the 'I' and the second 'I'. Below the letters is a thin red horizontal line.

Department of Applied Mathematics and Statistics

Department of Applied Mathematics and Statistics

Institute of Technology of Cambodia

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អត្ថបទនេះត្រូវបាន សេរសេរឡើងដោយដកកកថិកសកកកសកថ្យុកសកថ្មី្យក្សកលលលលលល
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ABSTRACT

LIST OF ABBREVIATIONS

GPUs	Graphics Processing Units
TPUs	Tensor Processing Units
API	Application Programming Interface
YOLO	You Only Look Once
JPG	Joint Photographic Experts Group
PNG	Portable Network Graphics
CSV	Comma Separated Values file
CNN	Convolutional Neural Network
VGG16	Visual Geometry Group 16
ResNet50	Residual Network 50
ReLU	Rectified Linear Unit
RGB	Red, Green and Blue
NSFW	Not Safe for Work
URLs	Uniform Resource Locators
mAP	Mean Average Precision

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I. INTRODUCTION

1.1 Presentation of Internship

This section formally documents the structured internship program, an essential academic component designed to bridge theoretical knowledge with professional practice in data science. The internship provides supervised, hands-on experience in applying data science methodologies to real-world challenges while developing both technical competencies and professional skills under the guidance of industry and academic mentors.

1.1.1 Objective of Internship

The program accelerates professional transition through mentored application of data science principles to organizational challenges, converting academic preparation into measurable workplace capabilities. This experiential learning opportunity serves as a critical transition from academic training to professional practice, ensuring that graduates meet industry standards and expectations.

1.1.2 Duration of Internship

Start date	03 August 2026
Duration	2 months
Location	
Position	Data Science Intern
Advisor	Mr. CHHEANG VINHA
Supervisor	Mr.
End date	30 October 2026

Table 1.1: Duration of Internship

1.2 Presentation of Organization

1.2.1 Vision

1.2.2 Mission

GDDE

1.2.3 Products And Services

- :
- :
- :
- :
- :

1.2.4 Organization Chart

1.2.5 Address and Contact

- Location:
- Email :
- Telephone : 0

1.3 Background

1.4 Problem Statement

1.5 Objectives

The main objective of this study is to identify and classify leaf diseases using Deep Learning technique. Specifically, this study aims to:

-
-
-

1.6 Scope and Limitation

This study focuses on application of DL techniques to identify tomato leaf disease. The dataset includes healthy and contains four classes of unhealthy tomatoes leaf disease. However, the study is subject to certain limitations, including:

-
-
-

1.7 Planning of Project

The following Gantt chart represents my scheduling during my internship at Kasegro for 2 months and fifteen days starting from March 04 to May 24, 2025. So, the details Gantt chart of the project is shown in the table below.

Task	Week											
	1	2	3	4	5	6	7	8	9	10	11	12
Explore sample dataset												
Research and Define the dataset for project												
Data Preprocessing												
CNN Design												
Transfer Learning												
Model Evaluation												
Deployment												
Internship Report												
Slide Presentation												

Table 1.2: Project Roadmap

II. LITERATURE REVIEW

Table 2.1: S

Source	Number	Methodology	Result

III. METHODOLOGY

3.1 Project Topology

3.2 Steps of Implementation

3.2.1 Data Collection

3.2.2 Preprocessing Data

3.2.3 Data augmentation

3.2.4 Train, Test and Validation

3.2.5 Model Selection

3.2.6 Performance Comparison

3.2.7 Experimental Setup

IV. RESULT AND DISCUSSION

4.1 Result

4.2 Deployment

V. CONCLUSION AND RECOMMENDATION

5.1 Conclusion

5.2 Recommendations

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APPENDICES